Research Article



Improving Enhancer Identification with a Multi-Classifier Stacked Ensemble Model

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Abstract

Enhancers are DNA regions that are responsible for controlling the expression of genes. Enhancers are usually found upstream or downstream of a gene, or even inside a gene's intron region, but are normally located at a distant location from the genes they control. By integrating experimental and computational approaches, it is possible to uncover enhancers within DNA sequences, which possess regulatory properties. Experimental techniques such as ChIP-seq and ATAC-seq can identify genomic regions that are associated with transcription factors or accessible to regulatory proteins. On the other hand, computational techniques can predict enhancers based on sequence features and epigenetic modifications. In our study, we have developed a multi-classifier stacked ensemble (MCSE-enhancer) model that can accurately identify enhancers. We utilized feature descriptors from various physiochemical properties as input for our six baseline classifiers and built a stacked classifier, which outperformed previous enhancer classification techniques in terms of accuracy, specificity, sensitivity, and Mathew's correlation coefficient. Our model achieved an accuracy of 81.5%, representing a 2–3% improvement over existing models.

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Introduction

Enhancers are short sequences of DNA located within the genome that regulate the expression of genes by modulating the rate of transcription.[1–4] These enhancers play a vital role in various biological processes, such as development, differentiation, and response to external stimuli. Enhancers can affect gene expression even at great distances, spanning hundreds of kilobases from the controlled gene.[5] They can be located upstream, downstream, or even within the coding region of a gene. Mutations to enhancer sequences can significantly impact gene expression. For example, a mutation that impedes the activity of an enhancer region can reduce gene expression, which, in turn, decreases the amount of protein produced. Conversely, a mutation that enhances the activity of an enhancer can increase gene expression, leading

to an overproduction of the protein that particular gene codes belong to.[6]

Mutations in enhancer sequences can significantly impact gene expression and potentially lead to diseases and abnormal development.[7] For instance, hereditary persistence of fetal hemoglobin (HPFH) is a condition in which individuals continue to produce fetal hemoglobin instead of converting it to adult hemoglobin after birth. HPFH offers some protection against inherited blood diseases such as Beta-thalassemia and sickle cell anemia. Similarly, mutations in enhancers that are involved in limb development can result in limb abnormalities, such as polydactyly, a condition where an individual has extra fingers or toes.

With the advent of high-throughput sequencing methods, researchers can now identify enhancers across the entire genome in various model organisms, including humans.[8] It is critical to understand the regulatory structure of the human genome and develop therapeutic approaches that specifically target enhancers in diseases such as cancer. This understanding depends on the identification of enhancers from the human genome.

Early research on enhancers has mostly been conducted through experimental methods, such as reporter assays and transgenic mouse models.[9] These methods have given researchers a better understanding of the regulatory components associated with specific genes and the functionality of enhancers. The activity of enhancers can be seen in cells using reporter assays, which enables researchers to gauge the level of activity and pinpoint potential constituents. Contrarily, using transgenic mouse models enables researchers to investigate the impact of enhancers in live things and see how they can affect the in vivo expression of a particular gene. These two methods offer crucial details about how enhancers operate and the procedures connected to them. The use of computational approaches has typically been in the discovery and classification of enhancers in order to better acknowledge and evaluate the huge genomics datasets that are becoming more and more common. This has been accomplished using a variety of techniques, including motif-based strategies, deep learning models, and machine learning algorithms.[10] These techniques have been crucial in assisting researchers to overcome the challenges given by analyzing and processing massive genomics datasets, allowing them to get insightful knowledge on the functional components of enhancers. Patterns in the data may be recognized using deep learning models, and enhancers can be categorized and analyzed using machine learning techniques and motif-based approaches. Researchers can better examine the complexity of these datasets and learn more about the regulatory functions of enhancers with the aid of computational tools. Motif-based techniques are useful for identifying potential enhancer regions by searching for specific DNA sequence motifs. These motifs are often known to be recognized by certain transcription factors. In recent years, machine learning methods such as random forest and support vector machines (SVMs) have been developed to classify enhancer regions based on their functional characteristics. This allows for a more accurate prediction of enhancer regions, as machine learning models can be trained on a range of genomic data to determine where functional enhancer elements are located. With the help of these machine learning methods, researchers to more precisely pinpoint possible enhancer sites and investigate their functional involvement in gene regulation. Both recurrent neural networks (RNNs) and convolutional neural networks (CNNs) have been applied to learn complex patterns from genomic data and to predict enhancer activity. While CNNs are best employed when learning patterns from data because they can discern complicated, non-linear relationships, RNNs are most appropriate for making predictions since they can account for the order of data points. RNNs employ feedback loops to store and analyze data continuously, while CNNs use convolutional layers to identify patterns in data. RNNs can be employed to forecast enhancer activity from a set of genomic data points, such as gene expression data or DNA sequences, by accounting for the order of the data points. CNNs can be utilized to learn complex patterns from genomic data, such as finding motifs or transcription factor binding sites, by recognizing detailed, non-linear connections between data points.

Both RNNs and CNNs have been used successfully to make predictions of enhancer activity and to learn intricate patterns from genomic data. By leveraging the strengths of each of these networks, researchers have been able to gain a better understanding of the regulatory networks of genomic data and to make more accurate predictions. Genomic techniques, including computational approaches based on machine learning and bioinformatics tools, as well as experimental techniques like chromatin immunoprecipitation followed by sequencing (ChIPseq).[11] ChIP-seq is a widely used experimental method for identifying enhancers, which involves using antibodies to extract chromatin fragments associated with transcription factors or other regulatory proteins. This technique has enabled researchers to identify enhancers and the regulatory proteins linked to them across the entire genome.[12] These methodologies have their own constraints and limitations, and there are still many areas of study that need to be explored. Using computational approaches based on machine learning and other bioinformatics technologies is another alternative for finding enhancers. Researchers, for example, have used epigenetic indicators such as DNA methylation patterns and histone changes to pin-

point the position of enhancers in the genome. Enhancers are often categorised according to their functional properties, such as tissue specialization, developmental stage specificity, or sensitivity to certain signaling pathways. Recent research has also concentrated on enhancer regulatory network properties, such as co-regulation by several transcription factors and the utilization of enhancer target gene connections to classify enhancers. These constraints in enhancer discovery and categorization have driven the development of novel techniques to address these issues. Integrating several types of data, such as epigenetic markers, chromatin accessibility, and transcription factor binding, for example, can increase enhancer identification accuracy and reliability.[13] Furthermore, the use of machine learning algorithms capable of handling large and complex datasets, such as deep neural networks, can improve classification performance.[14,15] Finally, the development of experimental techniques that can accurately measure enhancer function, such as massively parallel reporter assays (MPRAs), can help elucidate the mechanisms by which enhancers regulate gene expression and their alterations in disease. Overall, while there are still challenges in enhancer discovery and classification, continued advancements in both experimental and computational methods offer promising avenues for future research in this field. In this study, we developed a multi-classifier stacked-ensemble (MCSE) enhancer model for identifying enhancers. The methodology consists of a pipeline with various machine learning classifiers that take feature descriptors (DCC, PseKNC, SCPseDNC, PCPseDNC, NPCP, and kmer) as input. We trained the classifiers on a dataset of enhancers and non-enhancers. The performance of the classifiers was evaluated by measuring accuracy, sensitivity, specificity, and Mathew's correlation coefficient. The features were extracted using iLearn, a machine-learning tool. In order to take into account the fact that different machine learning classifiers have advantages with different types of data, we use voting and meta classifiers, where multiple classifiers are used as base classifiers. This is to boost our prediction performance and compare it to other approaches. Through this approach, we are able to compare our performance to other methods and increase our accuracy.

Dataset and Methods

The dataset used in this study consists of 200 bp sequences that correspond to enhancers. These sequences were compiled by Liu et al.[16] and were derived from nine different human genomic cell types, including H1ES, lymphoblast (K562), B-lymphocyte (GM12878), human hepatoma (HePG2), human umbilical vein endothelial cells (HUVEC), normal human lung fibroblasts (NHLF), and normal human epidermal keratinocytes (NHEK). A total of 2968 samples were obtained, consisting of 1484 enhancers and 1484 non-enhancers, of which 742 were strong enhancers and 742 were weak enhancers. This dataset was used for training and testing the enhancer classification model. Redundancy in the training and independent datasets was removed using the CD-HIT approach, which removed sequences with an average similarity of 80%. The resulting training dataset consisted of 1484 enhancer and 1484 non-enhancer samples, while the independent dataset consisted of 400 samples, including 200 enhancer and 200 non-enhancer sequences.

Methodology

The MCSE-enhancer model methodology involves different feature descriptors that are used to train different classifiers. Further, different classifiers are combined together for meta-learning (see Fig. 1).

Feature descriptors

The feature extraction process involves using a powerful and versatile Python-based toolkit called ilearn. This toolkit can extract and calculate 18 major sequence encoding schemes, encompassing 53 distinct types of feature descriptors. Additionally, ilearn can also be used to extract 6 major encoding schemes that cover 26 and 18 different types of feature descriptors for DNA and RNA sequences, respectively.

Dinucleotide-based Cross Covariance (DCC). The Dinucleotide-based Cross Covariance (DCC) encoding calculates the correlation between two physiochemical indices of dinucleotides separated by a certain number of nucleotides *lag* along the sequence. The formula for the DCC encoding is:

$$DCC(x_1, x_2, lag) = \sum_{i=1}^{L-lag-1} \left(T_{x_1}(Q_i)Q_{i+1} - \overline{T}_{x_1}\right) \times \left(T_{x_2}\left(Q_{i+lag}Q_{i+lag+1}\right) - \overline{T}_{x_2}\right)/(L - lag - 1) \quad (1)$$

where; x_1, x_2 = physiochemical indices, L = length of nucleotide sequence, for i^{th} position $T_{x_j}(Q_iQ_{i+1})$ represents the numerical measure of physiochemical index x_j for dinucleotide Q_iQ_{i+1} while $\overline{T}_{x_1}$ represents the average measure of physiochemical index x_j.

Feature Vector Dimension $= N * (N - 1) * lag$, where N = Number of Physiochemical indices and LAG = maximum of $lag(lag = 1, 2, \ldots\ldots lag)$.

Pseudo k-tupler Composition (PseKNC). PseKNC combines the advantages of both k-mer and composition-based descriptors, making it a more comprehensive feature extraction method.

PseKNC encoding scheme incorporate the k-tuple nucleotide composition which is given by:

$$D = \left[d_1, d_2, \ldots, d_{4^k}, d_{4^k+1}, \ldots, d_{4^k+\lambda}\right]^T \tag{2}$$

$$\frac{f_u}{\sum_{i=1}^{4^k} f_i + w\sum_{j=1}^{\lambda}\theta_j}, (1 \leqslant u \leqslant 4) \tag{3}$$

$$\frac{w\theta_{u-4^k}}{\sum_{i=1}^{4^k} f_i + w\sum_{j=1}^{\lambda}\theta_j}, \left(4^k \leqslant u \leqslant 4^k + \lambda\right) \tag{4}$$

where λ= total counted ranks or tiers of correlation along nucleotide sequence. $f_u\left(u = 1, 2, \ldots, 4^k\right)$ = frequency of nucleotide which is normalized to $\sum_{i=1}^{4k} f_i = 1$, w = factor and θ_j is defined as:

$$\theta_j = \frac{1}{L-j-1}\sum_{i=1}^{L-j-1}\Theta(R_iR_{i+1}, R_{i+j}R_{i+j+1}), (j = 1, 2, \ldots, \lambda; \lambda < L) \tag{5}$$

The correlation function $\theta\left(R_iR_{i+1}, R_{i+j}R_{i+j+1}\right)$ is given by:

$$\Theta(R_iR_{i+1}, R_{i+j}R_{i+j+1}) = \frac{1}{\mu}\sum_{v=1}^{\mu}\left[P_v(R_iR_{i+1}) - P_v\left(R_{i+j}R_{i+j+1}\right)\right]^2 \tag{6}$$

Parallel Correlation Pseudo Dinucleotide composition (PCPseDNC). PCPseDNC is based on the concept of Pseudo Dinucleotide Composition (PseDNC), which considers the correlations between adjacent dinucleotides in a DNA sequence. PseDNC has been shown to effectively capture the features of DNA sequences and is widely used in various bioinformatics applications. The difference between PCPseDNC and PseDNC is that PCPseDNC uses 38 default physiochemical indices instead of six physiochemical indices in PseDNC. It takes into account the parallel correlation between two dinucleotides at positions *i* and *j*, as well as the correlation between dinucleotides separated by a specific lag *k*.

K-mer. Kmer represents occurrence of frequencies of k-neighbouring acids which can be applied to human genomic regulatory sequence prediction. The 3-mer descriptor is calculated as:

$$f(t) = \frac{N(t)}{N}, t \in (AAA, AAC, AAG, .., TTT) \tag{7}$$

where, $N(t)$ = Number of Kmer type t, N = length of sequence

Series Correlation Pseudo Dinucleotide composition (SCPseDNC). Series Correlation Pseudo Dinucleotide composition (SCPseDNC) is a sequence encoding scheme that was proposed as an improvement over the existing methods for predicting RNA secondary structures. SCPseDNC considers the correlation between two different dinucleotides separated by a specific distance along the RNA sequence. It is an extension of the PCPseDNC method, which only considers parallel correlations.

SCPseDNC is defined as:

$$D = [d_1, d_2, \ldots\ldots..d_{16}, d_{16+1}, \ldots\ldots d_{16+\lambda}, d_{16+\lambda+1}, \ldots.., d_{16+\lambda\Lambda}]^T \tag{8}$$

where,

$$d_k = \frac{f_k}{\sum_{i=1}^{16}\left(f_i + w\sum_{j=1}^{\lambda}\theta_j\right)}, (1 \leqslant k \leqslant 16), \frac{w\theta_{k-16}}{\sum_{i=1}^{16}(f_i) + w\sum_{j=1}^{\lambda\Lambda}\theta_j}, (17 \leqslant k \leqslant 16 + \lambda\Lambda) \tag{9}$$

$f_k(k = 1, 2, \ldots..16)$= dinucleotide's normalized frequency occurrences. λ represents highest counted rank of correlation, w = weight factor ranged from 0 to 1, Λ = number of physicochemical indices. $\theta_j(j = 1, 2, \ldots, \lambda)$ is the j-tier correlation factor.

Nucleotide Physiochemical Properties (NPCP). The NPCP descriptor consists of 16 physiochemical properties that are calculated for each nucleotide, including electronic properties, hydrophobicity, steric properties, and polarizability. These properties are used to represent the chemical and physical properties of the nucleotide bases, such as the size, shape, and electronic charge distribution. To characterize a DNA sequence, NPCP implement a lone contribution including Zcurve, GC–Content, GC/AT skew, $(A+T)/(C+G)$ ratio which generates 3-, 1-, 1-, 2-dimensional vectors.

NPCP can be defined as:

$$NPCP(s_i) = Concatenate\left(f_j(s_i)(1 <= j <= 4)\right. \tag{10}$$

Here f_j is j^{th} property of NPCP.

MCSE-Enhancer model

The MCSE-enhancer model utilizes a stacked ensemble learning algorithm[17–19] that integrates six baseline classifiers, namely RF, ETC, LGBM, XGB, CBC, and MLP. During the model training phase, all classifiers are trained individually on combined feature descriptors, using KFold cross-validation, where 10% of the data is reserved for testing, another 10% is used for validation, and the remaining 80% is used for training. Each classifier is trained using a framework for hyperparameter optimization, which enables automatic hyperparameter adjustment and discovery of the search space that contains all possible values. Optuna is used to adjust hyperparameters such as the number of units and layers, activation functions, learning rate, and parameter regularization for the classifiers, resulting in improved performance.

Six different feature descriptors, varying in length and type, are extracted from DNA sequences, and they are merged before training each classifier independently. The best hyperparameters for each machine learning classifier are chosen, and their performance is evaluated on an independent dataset. The stacking ensemble learning algorithm uses k-fold estimation to map the training set (X, y) into a prediction set, Z, for each classifier (Z, y), ensuring that Z reflects test errors. A user-specified meta-learner is then used to predict y from Z. The MCSE-Enhancer model achieves an accuracy of 81.5% on an independent dataset, which is 3% higher than that of previous enhancer identification and classification frameworks.

Hyperparameter settings

Hyperparameters play a crucial role in machine learning as they determine the performance and behavior of the model. These settings are defined before the model is trained, and hyperparameter tuning is an important step in the machine learning process. It has a significant impact on the model's accuracy and ability to generalize. To optimize the performance of the MCSE-enhancer model, hyperparameter tuning was performed for all six baseline classifiers. A search space was defined for each classifier, which contained a range of possible values for various hyperparameters, such as regularization strength, learning rate, batch size, and the number of hidden layers.

Based on the literature and actual experience, the search space for each baseline classifier was determined. For example, the search space for the random forest classifier includes the number of trees, the maximum depth of the trees, the minimum number of samples required to split a node and others, while the search space for the LGB classifier includes the number of leaves, learning rate, maximum depth and others.

A combination of random and grid search algorithms were used to optimize the hyperparameters. The hyperparameter space was explored using a random search, and the top-performing hyperparameters were utilized as a starting point for a subsequent grid search. This iterative technique aided in effectively exploring the hyperparameter space and determining the best hyperparameter combination. The performance on the validation set during the KFold cross-validation phase was used to find the optimal combination of hyperparameters for each baseline classifier. The model's performance was assessed using a variety of criteria, including accuracy, specificity, sensitivity, and Mathew's correlation coefficient.

Table S1 in the supplementary file shows the appropriate hyperparameters for each baseline classifier. The hyperparameter settings were determined by the greatest accuracy obtained during the KFold cross-validation phase. Following that, these hyperparameters were used to train each classifier on the entire training set. The MCSE-enhancer model's performance was assessed on an independent test set, and the findings are provided in the preceding section. Overall, hyperparameter modification enhanced the model's performance dramatically, illustrating the importance of choosing ideal hyperparameters for machine learning algorithms.

Results and Discussion

Evaluation metrics

The accuracy (ACC), Matthew Correlation Coefficient (MCC), specificity (SP), and sensitivity (SN) matrices are used to evaluate performance in a classification task. For the identification of Enhancers, the results are divided into four categories: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{11}$$

$$Sensitivity = Recall = \frac{TP}{TP + FN} \tag{12}$$

$$Specificity = \frac{TN}{FP + TN} \tag{13}$$

$$MCC = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{14}$$

Sensitivity represents true positive samples that are classified as positive, specificity represents true negative samples that are classified as negative, TP is the total number of samples that are truly classified as positive, TN is the ratio of samples that are truly classified as negative, FP is the ratio of samples that are negative but classified as positive, and FN is the ratio of samples that are negative but classified as positive.

Performance between baseline classifiers

To further investigate the performance of the MCSE-enhancer model, we conducted a performance comparison between the six baseline classifiers used in our study. As shown in Figure 2, the performance of each individual feature descriptor was evaluated using the training dataset. The detailed data are shown in supplementary Table S2. The results demonstrate that each classifier performs differently based on the feature descriptor used. For example, the Catboost classifier achieves the highest accuracy of 77% on DCC, PseKNC, Kmer3, and NPCP feature descriptors, while LGBM achieves the highest accuracy of 77% on Kmer3 and NPCP feature descriptors. On the other hand, ETC achieves an accuracy of 76% on all feature descriptors, XGB obtains an accuracy of 76% on DCC, PseKNC, and Kmer feature descriptors, RF

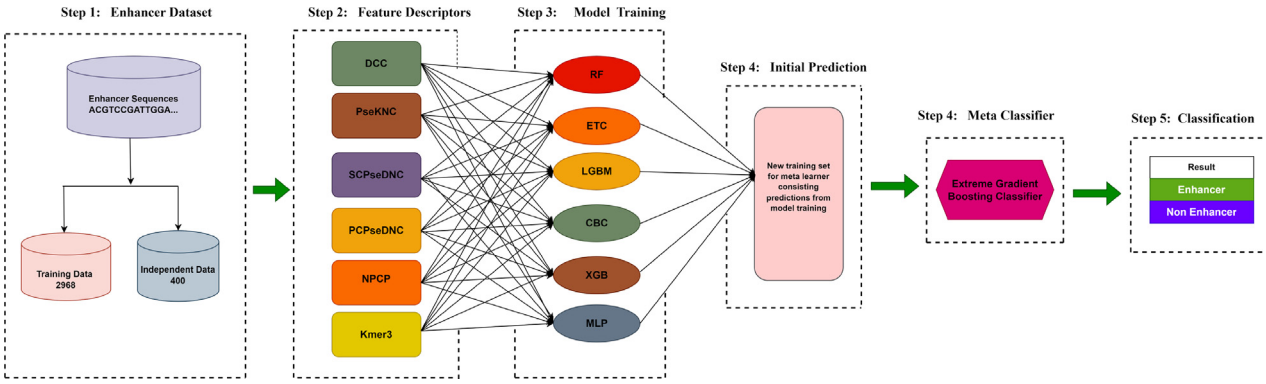


Figure 1. Shows the MCSE-Enhancer model framework which is summarized in following steps: (1) data collection, (2) Feature extraction, (3) model construction, (4) stacking of baseline models, (5) classification.

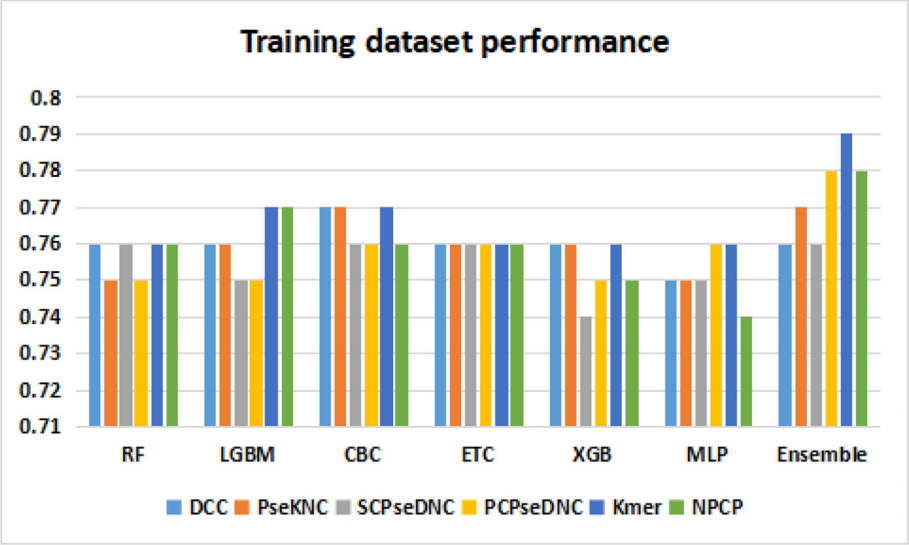


Figure 2. The enhancer identification performance comparision between six baseline classifiers on every individual descriptor using training dataset.

achieves an accuracy of 76% on DCC, Kmer3, and NPCP feature descriptors, and MLP achieves an accuracy of 76% on PCPseDNC and Kmer feature descriptors.

To assess the classifiers' performance on an independent dataset, we showed the accuracy of each classifier for each feature descriptor in Figure 3. The detailed data are shown in supplementary Table S3. The results demonstrate that the Catboost and Extra-tree classifiers achieve the highest accuracy of 81% on the DCC feature descriptor, while Multilayer perceptron obtains the highest accuracy of 80% on the NPCP feature descriptor. Interestingly, the ensemble mechanism used in our MCSE-Enhancer model outperforms individual classifiers, which is evident from the higher accuracy achieved on the independent dataset. The stacked ensemble model integrates the strengths of the individual classifiers and results in a more robust and accurate model for enhancer identification.

Effects of stacking ensemble learning algorithm

Figure 4 shows the performance comparison of the different classifiers and the ensemble mechanism in terms of accuracy on an independent dataset. the detailed data is shown in the supplementary Table S4. The ensemble mechanism outperformed all the individual classifiers, achieving higher accuracy. This improvement in accuracy demonstrates the effectiveness of the ensemble mechanism in enhancing the overall performance of the model.

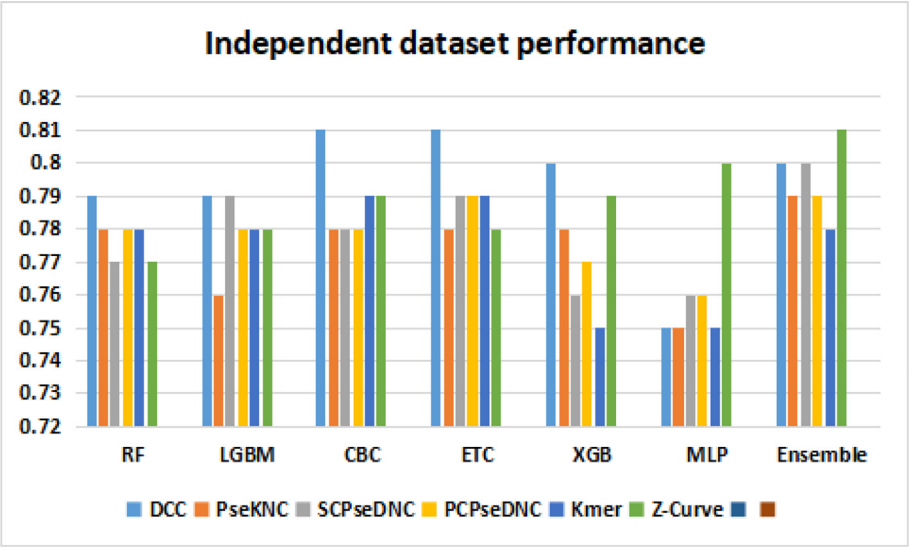


Figure 3. The enhancer classification performance comparision between six baseline classifiers on every individual descriptor using independent dataset.

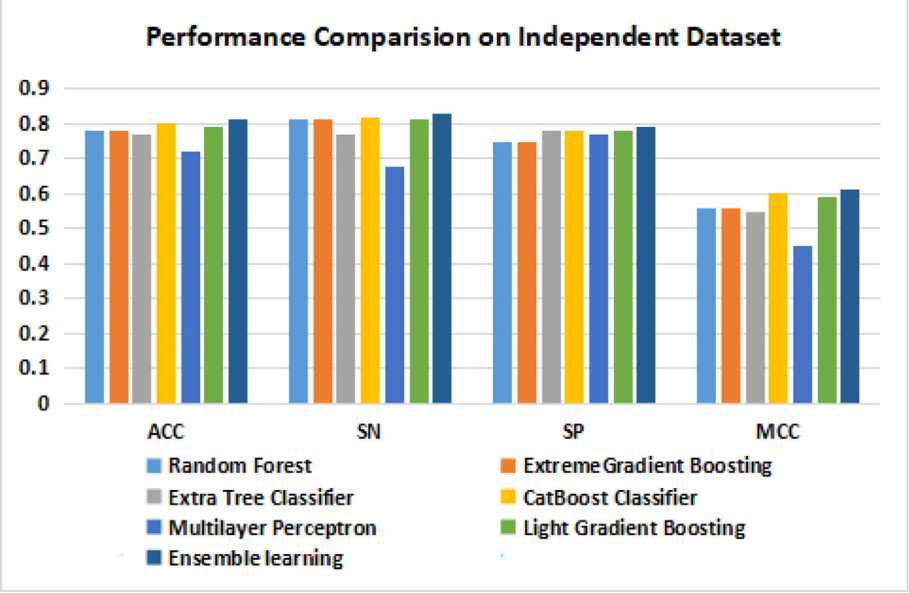


Figure 4. The performance comparison of the different classifiers and the ensemble mechanism in terms of accuracy on an independent dataset.

Stacking is indeed an ensemble learning technique that can help improve the performance of predictions by combining multiple models. Unlike traditional ensemble methods, such as averaging or voting, stacking uses a meta-model, which is trained to learn how to combine the predictions of the base models. The meta-model takes the predictions of the base models as input and learns how to weigh or adjust them to generate a final prediction that is more accurate than any of the base models alone.

The ensemble mechanism's better performance can be due to its capacity to integrate the strengths of numerous classifiers while limiting their limitations. The stacking ensemble approach employed in this work combines the predictions of the six baseline classifiers by learning the link between the individual classifier predictions and the true class labels using a meta-learner. The ensemble technique improves the overall performance of the model by using the varied range of feature descriptors collected from the

DNA sequences and the individual capabilities of each classifier. According to the results presented in Figure 4, the stacking ensemble learning method achieved an accuracy of 81.5%, specificity of 0.79%, sensitivity of 0.83%, and MCC of 0.59. These performance metrics are all better than those obtained by the individual base models, indicating that the stacking technique was able to effectively combine the strengths of the individual models and produce a more accurate and reliable prediction.

Furthermore, the ensemble process reduces the risk of overfitting by mapping the training set into a prediction set for all classifiers using k-fold estimation followed by using a meta-learner to forecast the true labels of classes from the merged predictions. This decreases the possibility of anyone classifier or feature descriptor having an undue effect on the model, resulting in a more resilient and accurate model. Overall, the ensemble technique utilized in this work is a successful method for enhancing the accuracy of enhancer identification models by utilizing the strengths and minimizing the flaws of various classifiers.

Analysis of different descriptor

We used SHAP analysis in our MCSE-Enhancer model to identify the relative relevance of each descriptor. The SHAP analysis determines how much each descriptor contributes to the final forecast. The descriptor SCPseDNC contributed the most to the final prediction, as shown in Figure 5, with a weight of 46.06%. This is followed by DCC and PseKNC, each with a 9.5% contribution, and kmer, with a 9.91% contribution. NPCP, on the other hand, made the smallest contribution of only 0.08%.

According to the findings, PCPseDNC is the most informative descriptor for identifying enhancers, followed by SCPseDNC Kmer, PseKNC and DCC. These findings imply that these descriptors capture critical information about enhancer areas and can aid in enhancer prediction accuracy. The low contribution of NPCP shows that this descriptor may be less useful in finding enhancers, which is consistent with earlier research. The importance of each individual feature descriptor is given in supplementary file. i.e. (Figure S1 shows SHAP analysis of DCC, figure S2 shows SHAP analysis of PseKNC, figure S3 shows SHAP analysis of SCPseDNC, figure S4 shows SHAP analysis PCPseDNC, figure S5 shaows SHAP analysis of Kmer, and figure S6 shows SHAP analysis of NPCP) feature descriptors.

Overall, our SHAP analysis sheds light on the relative relevance of each descriptor in MCSE-Enhancer model and may be used to guide future research on enhancer prediction.

Comparison with existing enhancer identification methods

To further evaluate the effectiveness of our MCSE-Enhancer model, we compared it with other models on a different test dataset, including iEnhancer-2L,[20] iEnhancer-XG,[11] Enhancer-Pred,[21] iEnhancer-ECNN,[12] iEnhancer-EL,[13] Enhancer-IF,[14] iEnhancer-EBLSTM,[15] and BERT-2DCNNs.[22] iEnhancer-2L[20] is a two-layer classifier that uses the first layer to determine whether a sequence is an enhancer or not and the second layer to determine the strength of the sequence. Enhancer-Pred utilizes SVM models for corresponding prediction model construction.

The ensemble learning concept is used by iEnhancer-EL to create a two-layer ensemble classifier. iEnhancer-XG employs XGBoost[17] on five physiochemical characteristics to create a two-layer identification model. iEnhancer-ECNN

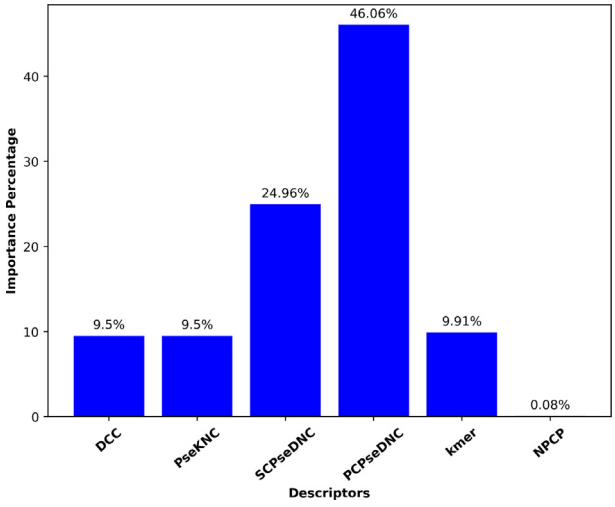


Figure 5. Importance of different feature descriptors in enhancer identification, computed using SHAP analysis.

Table 1 Results of each model on independent test dataset.

Methods	ACC(%)	MCC	SN(%)	SP(%)
MCSE-Enhancer	**81.5**	**0.59**	**83**	79
iEnhancer-2L	73	0.460	70	75
iEnhancer-XG	75.75	0.515	74	77.5
Enhancer-Pred	74	0.480	73.5	74.5
iEnhancer-ECNN	76.9	0.537	78.5	75.2
iEnhancer-EL	74.75	0.496	71	78.5
Enhancer-IF	79.3	0.585	80.5	78.78
iEnhancer-EBLSTM	77.2	0.534	75.5	**79.5**
BERT-2D CNN	75.6	0.514	80	71.2

and iEnhancer-EBLSTM use deep learning techniques to identify enhancers. BERT-2DCNNs creates a 2D CNN model using features taken from previously trained BERT models. Enhancer-IF utilizes five base classifiers to determine the cell-specificity of enhancers on eight different cell-lines. The results of the comparison analysis are displayed in Table 1. In comparison to other approaches, MCSE-Enhancer model outperformed them in terms of ACC, MCC and sensitivity. However the proposed model was unable to achieve the best specificity.

Conclusion

In the study, "Improving Enhancer Identification with a MCSE-Enhancer Model" introduces a novel approach to enhancer discovery using a stacked ensemble framework. The approach combines various machine learning models trained on different feature descriptors, resulting in improved prediction performance compared to other cutting-edge methods. The results of the study demonstrate the potential of this methodology in enhancer discovery, with higher accuracy, Mathews correlation coefficient, sensitivity, and specificity. The study highlights the significance of ensemble learning methods in improving DNA sequence analysis precision and presents a useful framework that could be applied to solving other relevant genomics problems. Future work could further improve the performance of the ensemble framework by incorporating additional machine-learning models or exploring different feature extraction strategies. A web server of the MCSE-Enhancer is made freely available at:https://nsclbio.jbnu.ac.kr/tools/MCSE-Enhancer/.

Data Availability

The dataset and code is freely available at: https://github.com/MirBilalAhmad/DNA-Enhancers.

CRediT authorship contribution statement

Bilal Ahmad Mir: Conceptualization, Investigation, Writing – original draft, Methodology, Writing – review & editing. **Mobeen Ur Rehman:** Conceptualization, Investigation, Writing – original draft, Methodology, Writing – review & editing. **Hilal Tayara:** Conceptualization, Supervision, Writing – original draft, Project administration, Methodology, Funding acquisition. **Kil To Chong:** Conceptualization, Supervision, Project administration, Methodology, Funding acquisition.

DATA AVAILABILITY

Data will be made available on request.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary Data

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